(18)

↳ % change :-

→ Creating formula for % change :-

Ex:- 30 is changed to 36 then % change ?

[By using technique of basic]

30 is initial value ⟶ 100% | 36 as final value

30 → 100%
3 → 10%
6 → 20%

30 + 3 + 3 = 100% + 10% + 10% = 120%

120% − 100% = 20% change.

(or)

30 → 100%
3 → 10%
6 → 20%

change b/w 36 & 30 in integers
36 − 30 = 6
But we want in terms of '%'.

* change in value is always compared with respect to initial value.

→ Step 1:- Observe diff b/w final & initial (integer value as result)
[36 − 30 = 6.]

Step 2:- compare this change w.r.t initial. $\left[\frac{36-30}{30}\right]$

Step 3:- Convert this fractional value into % [×100]

$$\% \text{ change} = \frac{36-30}{30} \times 100 = +20\% \quad [+ \text{ means } \uparrow^{ed}]$$

$$\boxed{\text{Formula} = \% \text{ change} = \frac{\text{final} - \text{initial}}{\text{initial}} \times 100}$$

↳ ↑/↓ in % to original value :- [By converting into decimal]

$$100\% = \frac{100}{100} = \frac{1}{\text{fraction}} = 1.0 \quad (\text{decimal})$$

if 10% ↑ ⇒ 100% + 10% = 110% $= \frac{110}{100} = 1.1$

$= 1 + 0.1$ [Means when 10% ↑ed resultant is 1+0.1]

If 20% ↑ed ⇒ $\frac{100+20}{100} = 1.20$

(or)

$20\% = \frac{20}{100} = 0.2$ ↑ed to 1 = 1 + 0.2 = 1.2

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→ If 30% ↓ed ⇒ 100% − 30% = 70% = $\frac{70}{100}$ = 0.7

(or)

30% = $\frac{30}{100}$ = 0.3 sub from 1 (100%) = 1 − 0.3 = 0.7

Conclusion: ① If ↓ed respectable decimal value subtracted from 1 to get resultant factor.

② If ↑ed respectable decimal value added to 1 to get resultant decimal factor.

Ex:- 40% ↑ed to 2500 then resultant Rs ?

i) 2500 → 100%
250 → 10%
4×250 = 1000 → 40%

100% + 40% = 2500 + 1000 = 3500

(or) (or) $\left[1(100\%)+(40\%)0.4\right]2500$
$= 1.4 \times 2500 = 3500$

↳ Net change when 2 (or) more decreases (or) % one after the other:- ↓ed / ↑ed.

Ex:-1:- 10000 Rs (i) ↓ 20% (ii) ↑ 10% (iii) ↓ 30% then result ?

10000
−2000 — 20% of 10000 = 2000
8000 = (1 − 0.2) 10000 = 0.8 × 10k
+800 — 10% of 8k = 1k
8800 = [(1 + 0.1) 8000 = 1.1 × 8000 = 1.1 × 0.8 × 10k
− 2640 = 30% of 8800 = 3 × 880 =
6160 = (1 − 0.3) 8800 = 0.7 × 8800 = 0.7 × 1.1 × 0.8 × 10k.

Ex:-2:- 25000 (i) ↑ 20% (ii) ↓ 30% [2 times only change]

directly Resultant Rs = (1 + 0.2)(1 − 0.3) 25000
= 1.2 × 0.7 × 25000 = 21000

(or)

* Net change (or) % when (i) ↑/↓ by x% (ii) ↑/↓ by y% Rs

↑ → +, ↓ → −

① x% ↑, y% ↑ → Net change = $\left[+x+y+\frac{xy}{100}\right]$%

② x% ↑, y% ↓ → $\left[+x-y-(+x-)\frac{xy}{100}\right]$%

③ x% ↓, y% ↑ → $\left[-x+y-\frac{xy}{100}\right]$%

④ x% ↓, y% ↓ → $\left[-x-y+(+x-)\frac{xy}{100}\right]$%

↳ 25000 changed (i) 20% ↑ (ii) 30% ↓ed

$+x, -y \rightarrow \left[+20-30-\frac{20\times30}{100}\right]\% = -10-6 = -16\% =$ Net change

Result = 25000 (100%) − 16% (16% of 25000)

= 25000 − 4000 = 2100

✱ Net change when $\pm x\%, \pm y\% = \left[\pm x\% \pm y \pm \frac{xy}{100}\right]\%$

↳ Purpose of conversion table:-

Ex:- 33.33% of 9000 + 37.5% of 8000 + 42.84% of 49000 = ?

$\frac{1}{3}\times9000 + \frac{3}{8}\times8000 + \frac{3}{7}\times49000 = 3000+3000+21000 = 27000$

↳ Purpose of ladder Rule:-

Ex:- the price of petrol is ↑ed by 25%, then how much consumption to be reduced to maintain same expenditure?

Assume Initial → 100%

25% ↑ 25%

125%

to make same expenditure after ↑ed by 25% the value 125% ↓ed to 100%. Means % change from 125% to 100%

$= \frac{100\%-125\%}{125\%}\times100 =$ −20% ↓ed

$= \frac{25\%}{125\%}\times100$

20% consumption to be reduced.

Creation of formula:- If price ↑/↓ by x%, the % consumption to be ↓ed/↑ed as.

Price ↑ed then ↓ed consum $= \frac{x}{100+x}\times100\%$

Price ↓ed then ↑ed qn cons % $= \frac{x}{100-x}\times100\%$

(21)

By using ladder rule:-

Price ↑25% then consum ↓% = ?

